Minia University

Faculty of Engineering

Mechatronics & Industrial Robotics Program

Electronic Circuits Report
Proteus Simulation Tasks

Course:

Electronic Circuits

Submitted to:

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Academic Year

2025 – 2026

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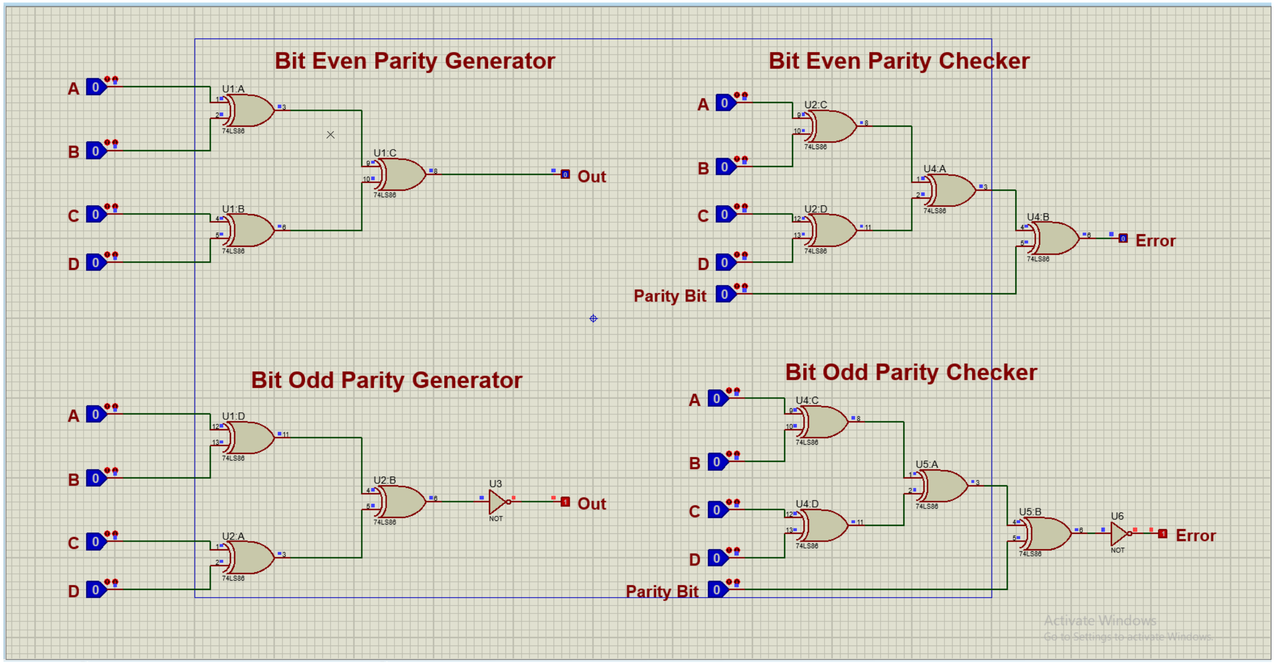
Task 01: Parity Bit Circuit

Muhammed



Objective:

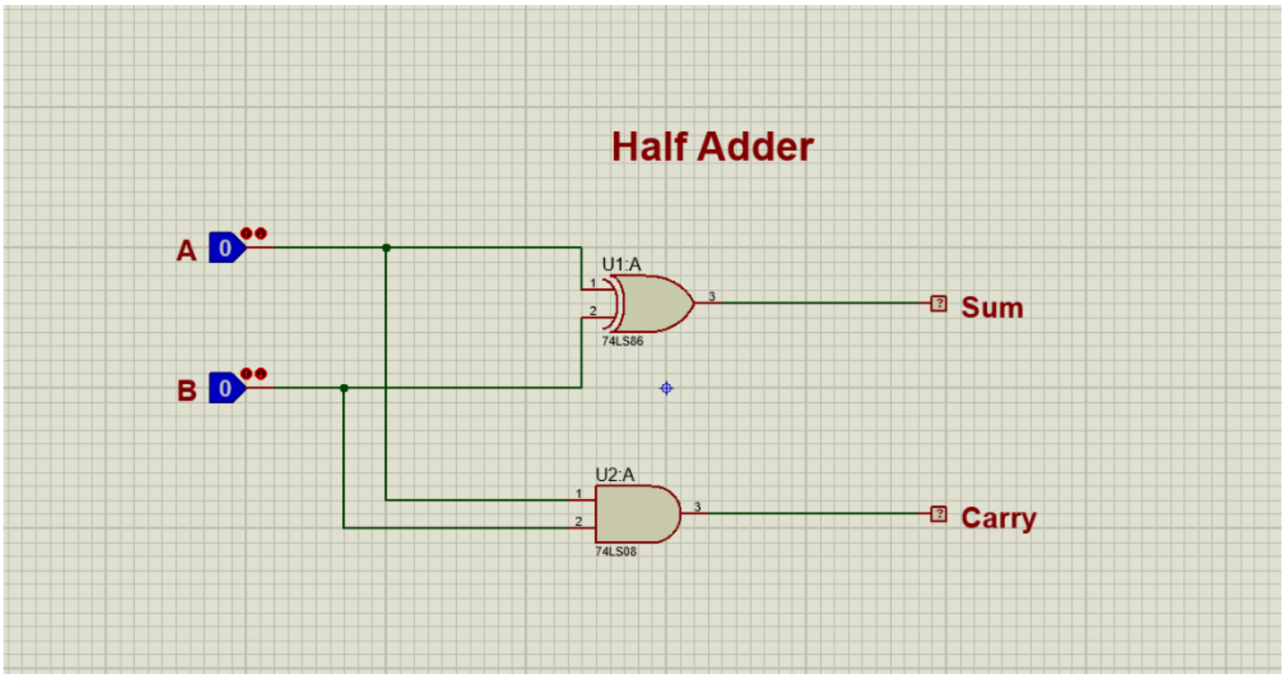
To design and simulate a 4-input parity bit generator using logic gates.

Theory:

A parity bit is used for error detection in digital communication. In even parity, the total number of 1s must be even.

Task 02: Half Adder Circuit

Muhammed



Objective:

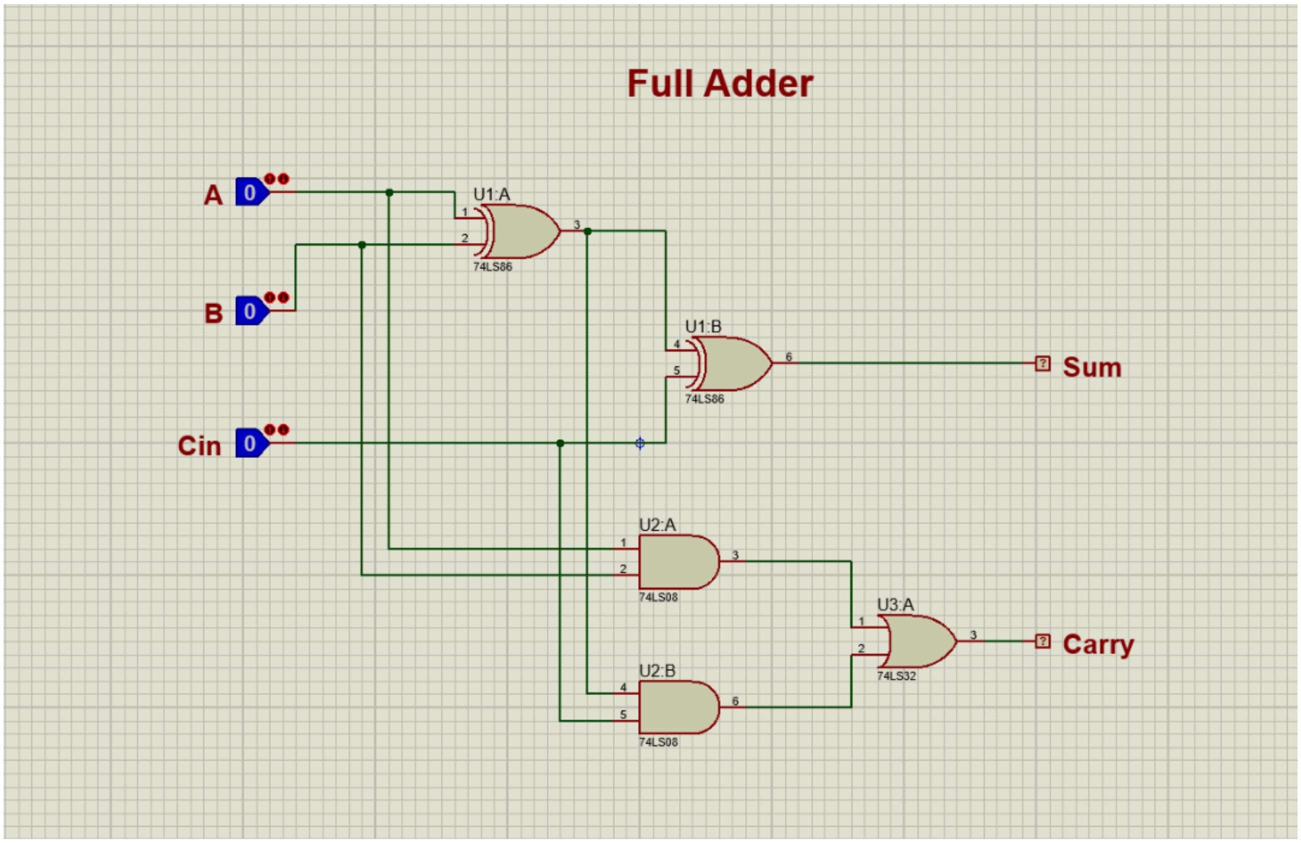
To design and simulate a half adder circuit using logic gates.

Theory:

A half adder is a combinational logic circuit used to add two single-bit binary numbers. It produces two outputs: Sum and Carry.

Task 03: Full Adder Circuit

Muhammed



Objective:

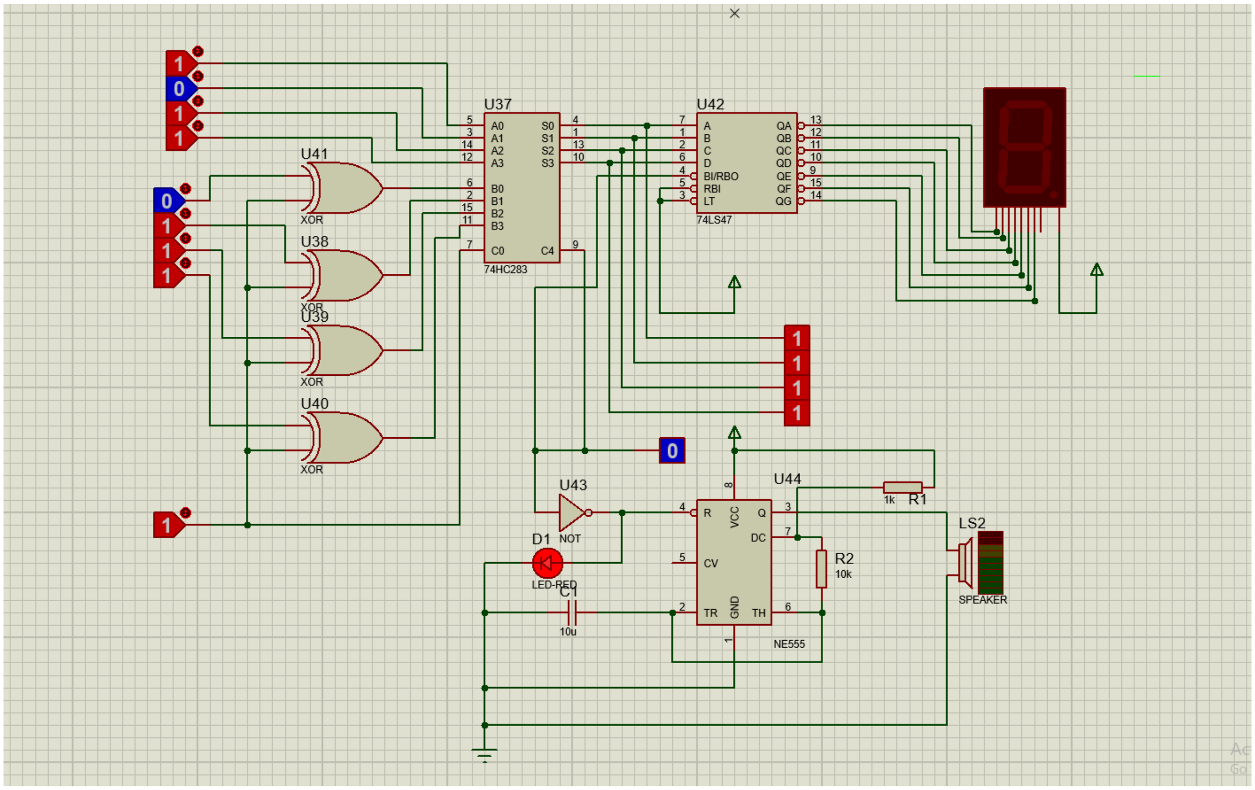
To design and simulate a full adder circuit using logic gates.

Theory:

A full adder is a combinational logic circuit used to add three binary inputs: A, B, and Cin. It produces two outputs: Sum and Cout.

Task 04: Sub adder Circuit and Applications

Gerges



Objective:

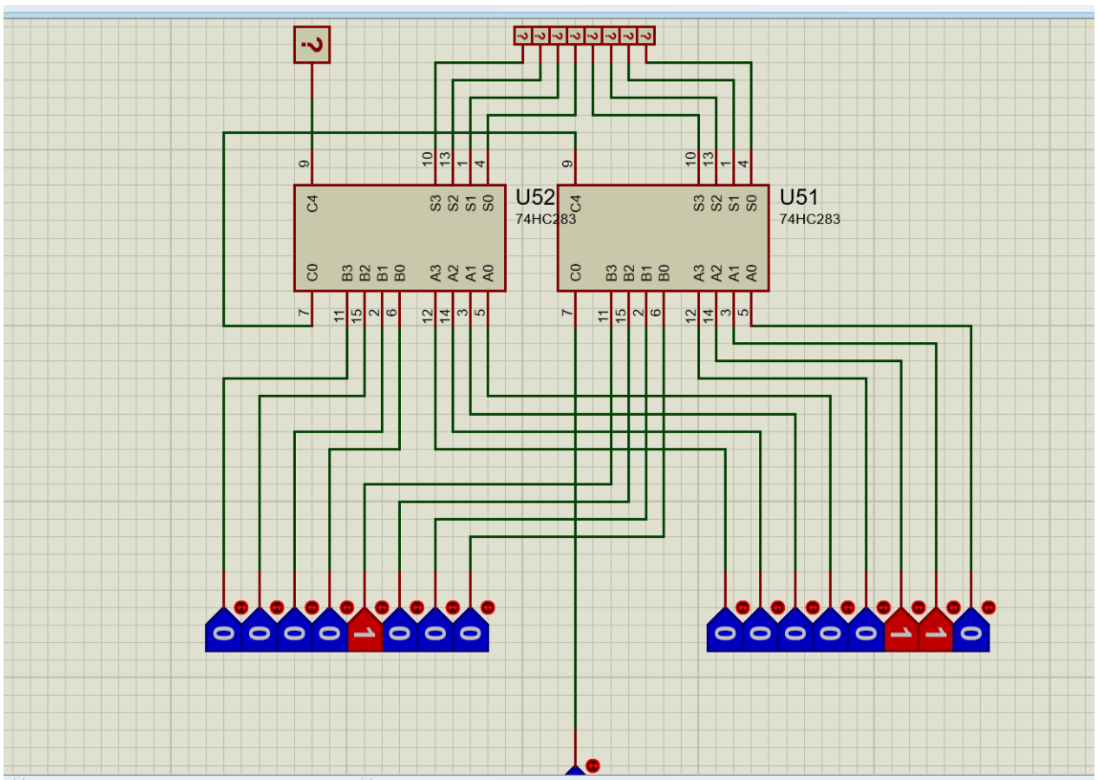
To design and simulate a 4-bit subtractor circuit with a negative result alarm.

Theory:

The circuit uses the 2’s complement method to perform subtraction. If the subtraction result is negative, the 7-segment display is turned off and an alarm is activated using a red LED and sound circuit.

Task 05: 8-Bit Adder Using 74HC283

Gerges



Objective:

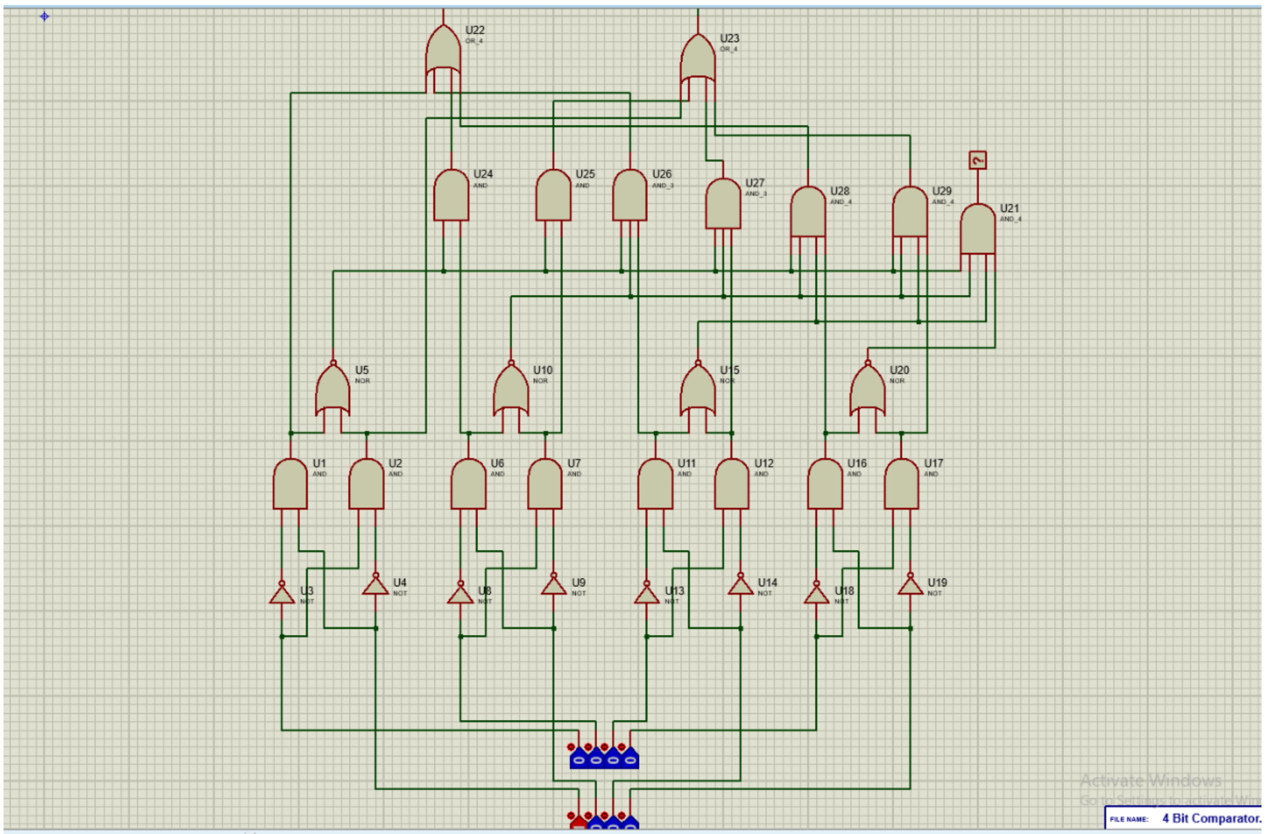
To design and simulate an 8-bit binary adder using two 74HC283 4-bit adder ICs.

Theory:

The 8-bit addition is achieved by cascading two 4-bit 74HC283 adders, where the carry output of the lower 4-bit adder is connected to the carry input of the higher 4-bit adder.

Task 06: 4-Bit Comparator from Scratch

Gerges



Objective:

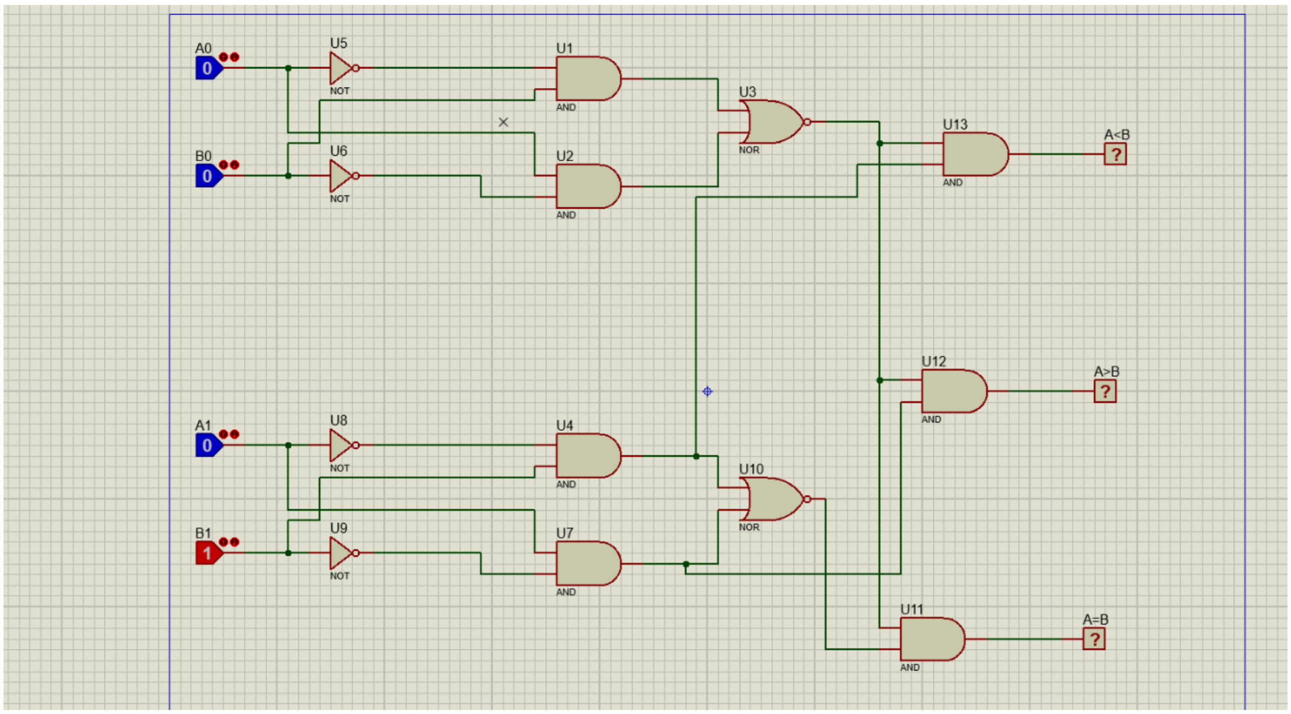
To design and simulate a 4-bit magnitude comparator using basic logic gates.

Theory:

The circuit compares two 4-bit binary numbers and determines whether A is greater than, equal to, or less than B.

Task 07: 2-Bit Comparator Circuit

Habiba



Objective:

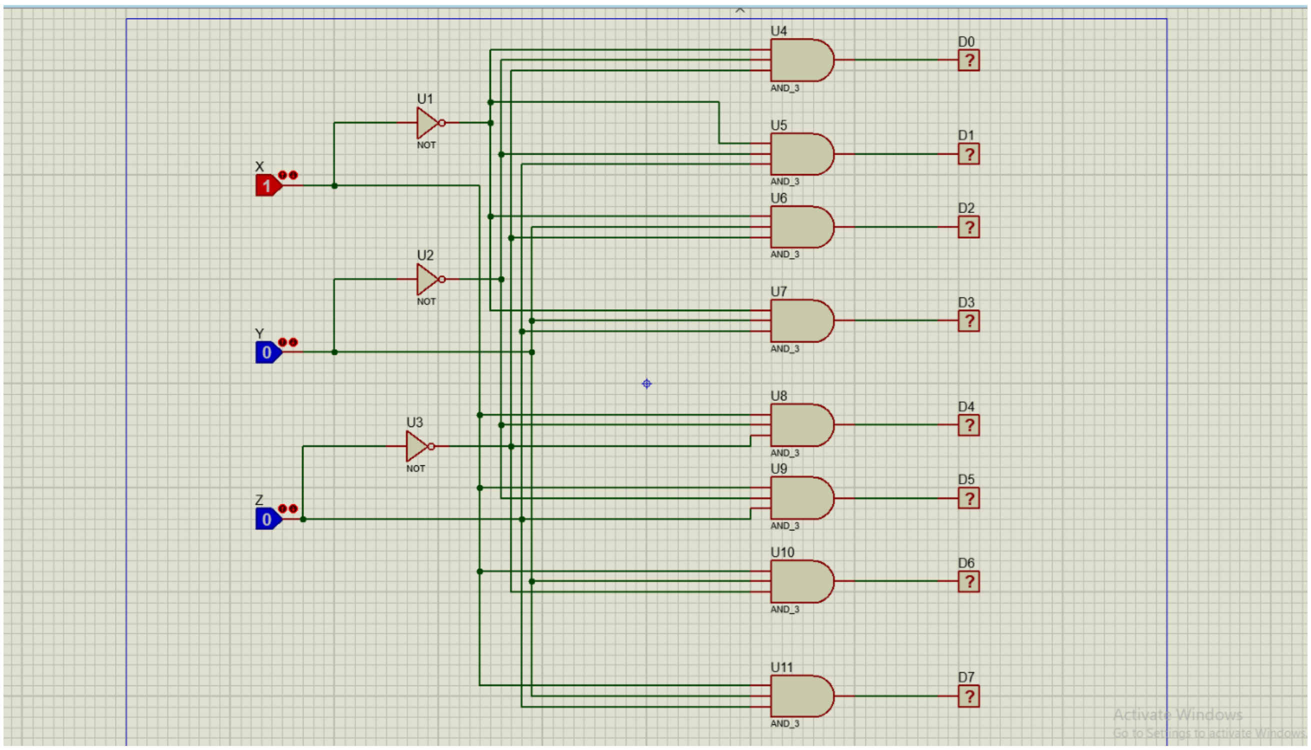
To design and simulate a 2-bit comparator circuit using basic logic gates.

Theory:

The circuit compares two 2-bit binary numbers and gives one of three outputs: A > B, A = B, or A < B.

Task 08: Decoder Circuit

Habiba



Objective:

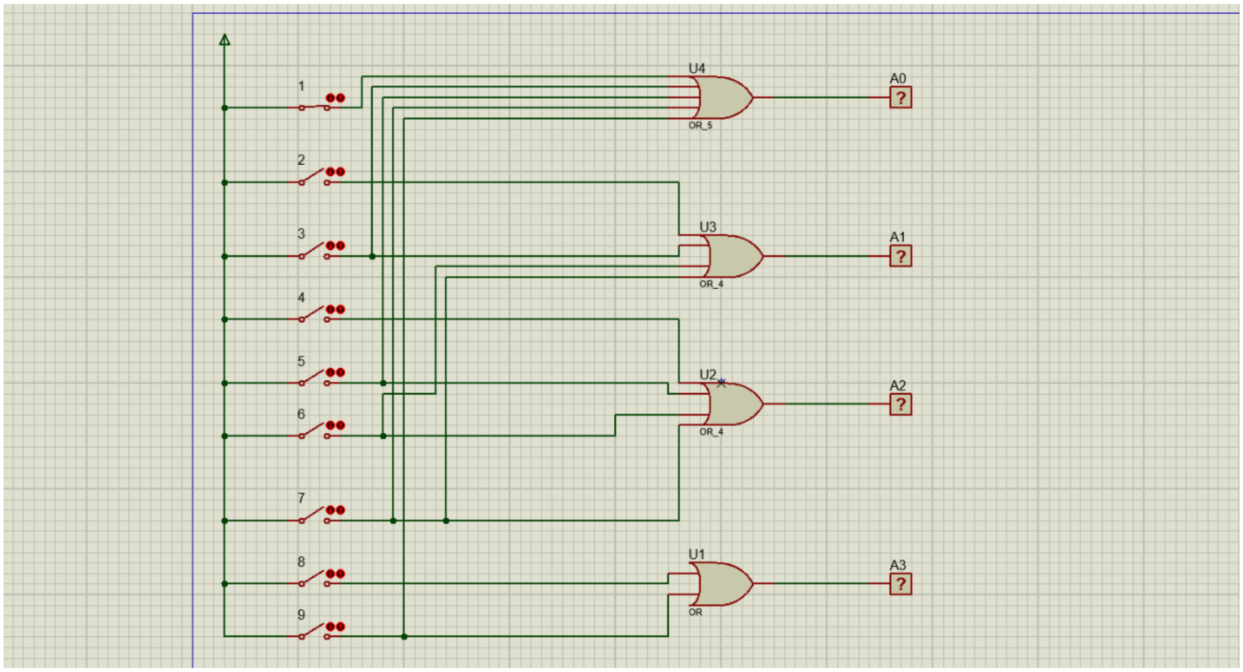
To design and simulate a decoder circuit using basic logic gates.

Theory:

A decoder converts binary input lines into one active output line based on the input combination.

Task 09: Encoder Circuit

Habiba



Objective:

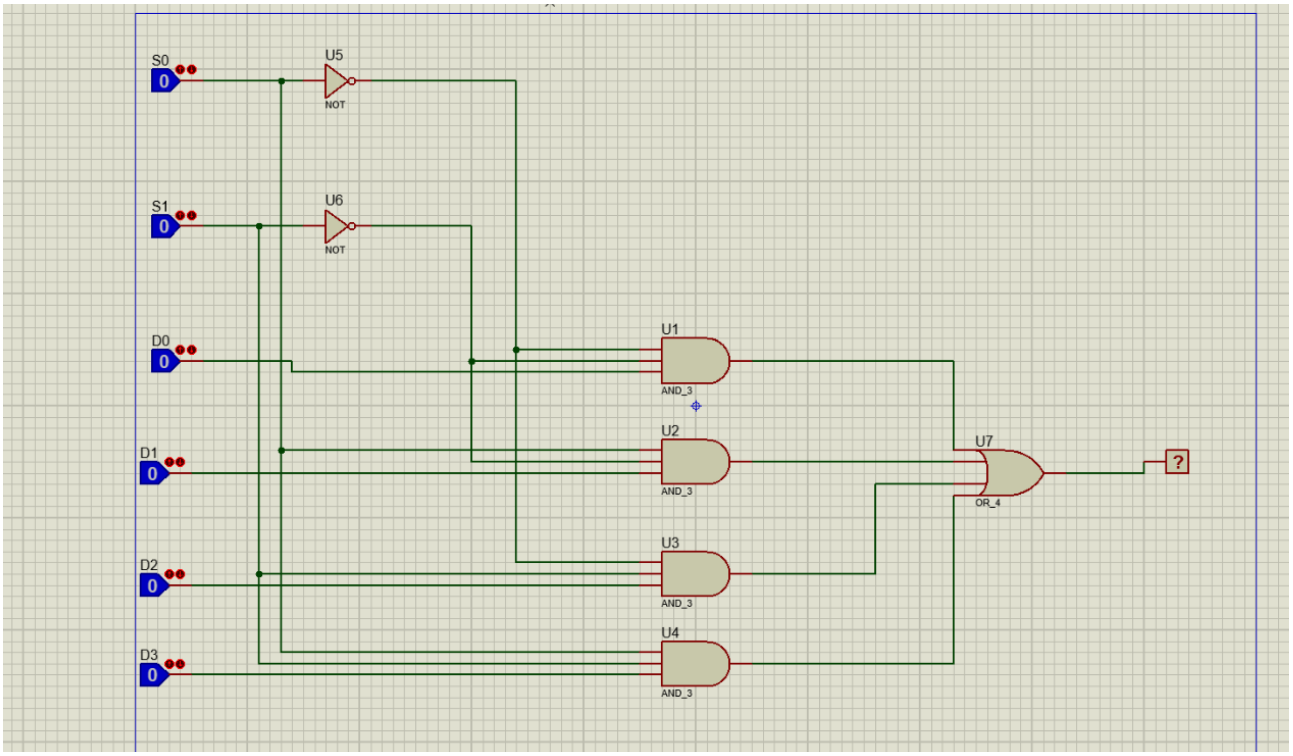
To design and simulate an encoder circuit using basic logic gates.

Theory:

An encoder converts one active input line into a binary output code that represents the active input.

Task 10: Multiplexer Circuit

Gehad



Objective:

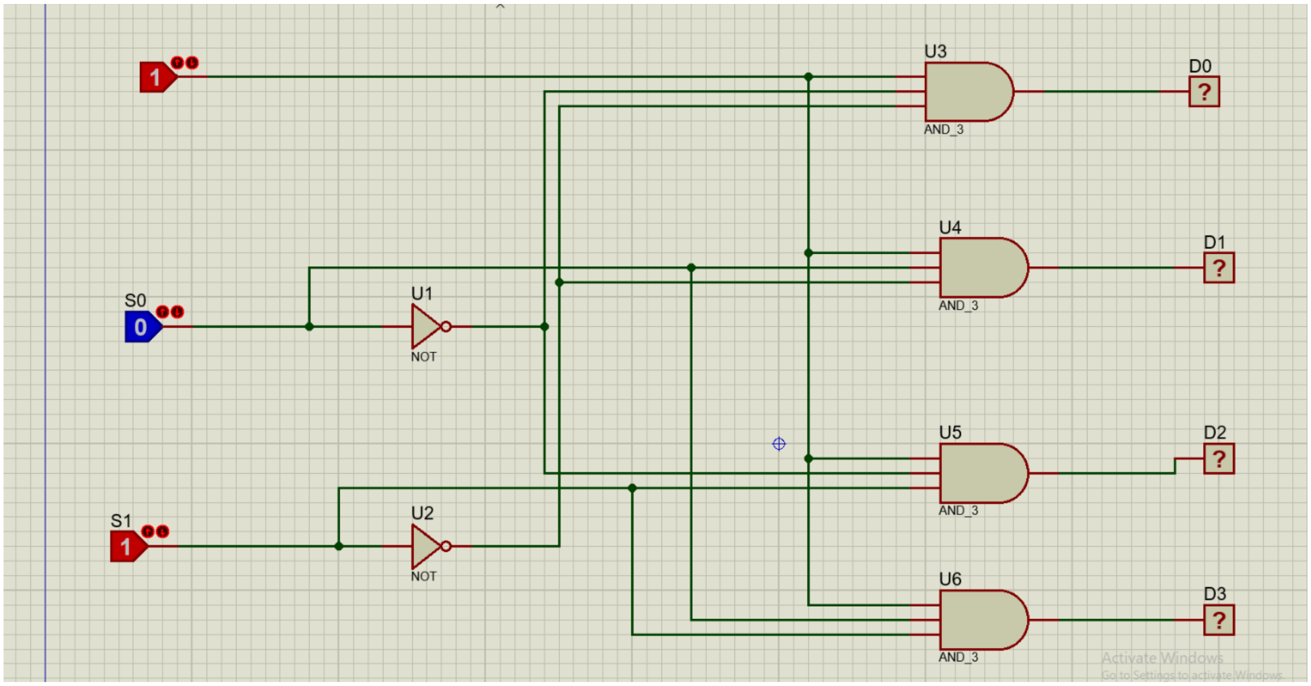
To design and simulate a multiplexer circuit using basic logic gates.

Theory:

A multiplexer selects one input from several input lines and passes it to a single output according to the select lines.

Task 11: Demultiplexer Circuit

Gehad



Objective:

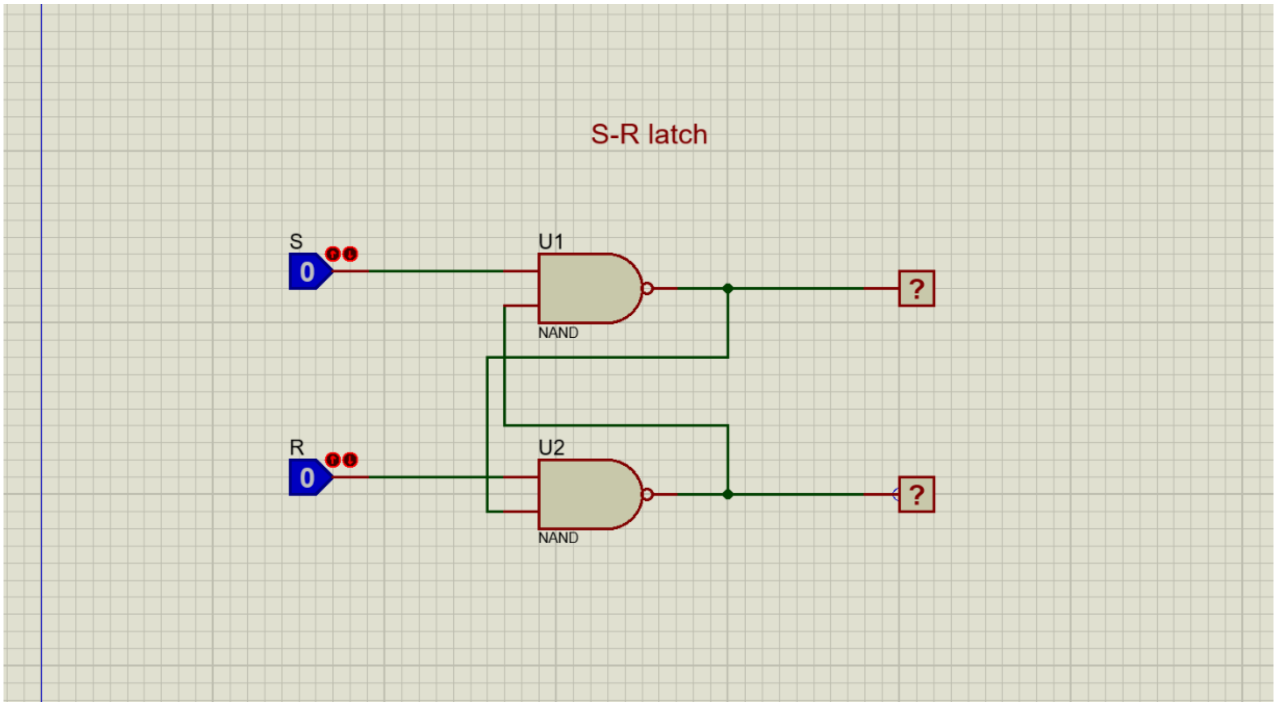
To design and simulate a demultiplexer circuit using basic logic gates.

Theory:

A demultiplexer takes one input signal and sends it to one of several output lines according to the select lines.

Task 12: S-R Latch Circuit

Gehad



Objective:

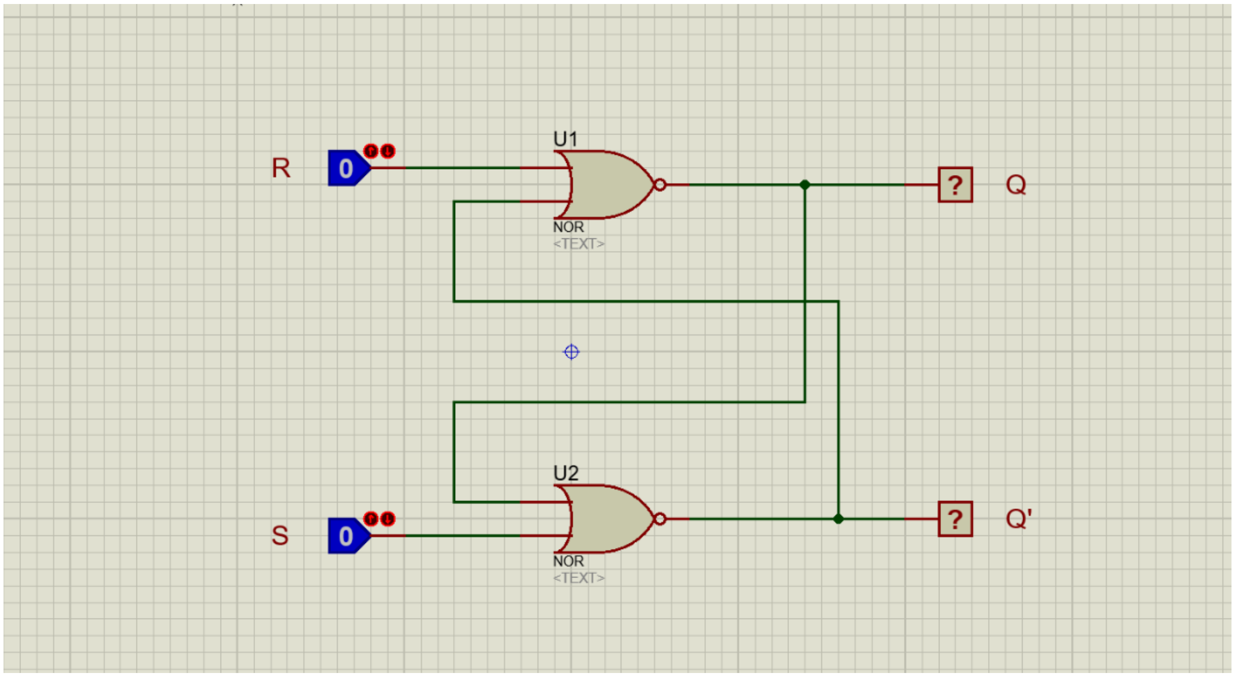
To design and simulate an S-R latch circuit using NAND gates.

Theory:

An S-R latch is a basic memory circuit that stores one bit of data using Set and Reset inputs.

Task 13: R-S Latch Circuit

Mohrael



Objective:

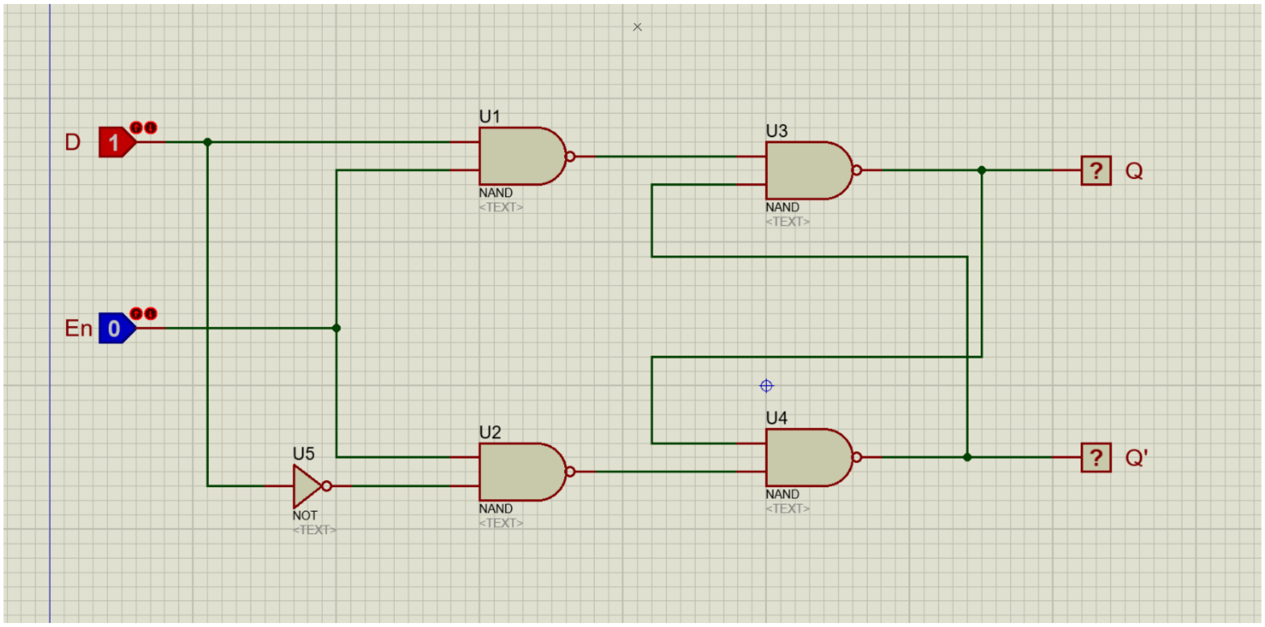
To design and simulate an R-S latch circuit using basic logic gates.

Theory:

An R-S latch is a basic memory circuit that stores one bit of data using Reset and Set inputs.

Task 14: D Latch Circuit

Mohrael



Objective:

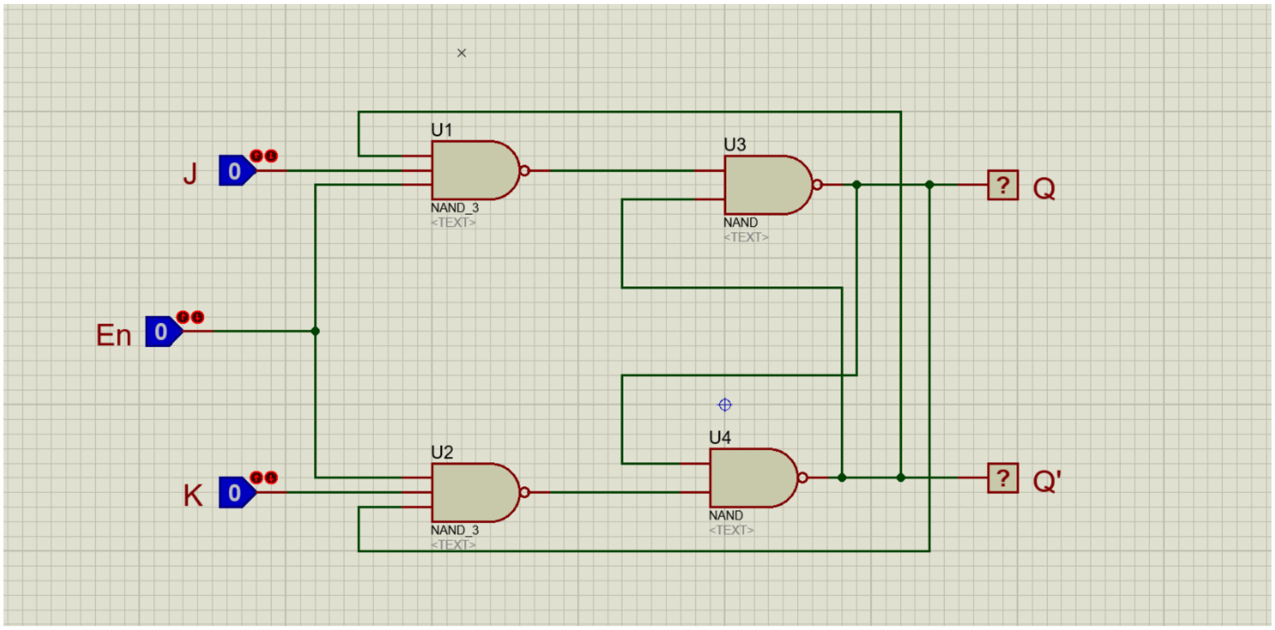
To design and simulate a D latch circuit using basic logic gates.

Theory:

A D latch stores the value of the data input D when the enable input is active.

Task 15: J-K Latch Circuit

Mohrael



Objective:

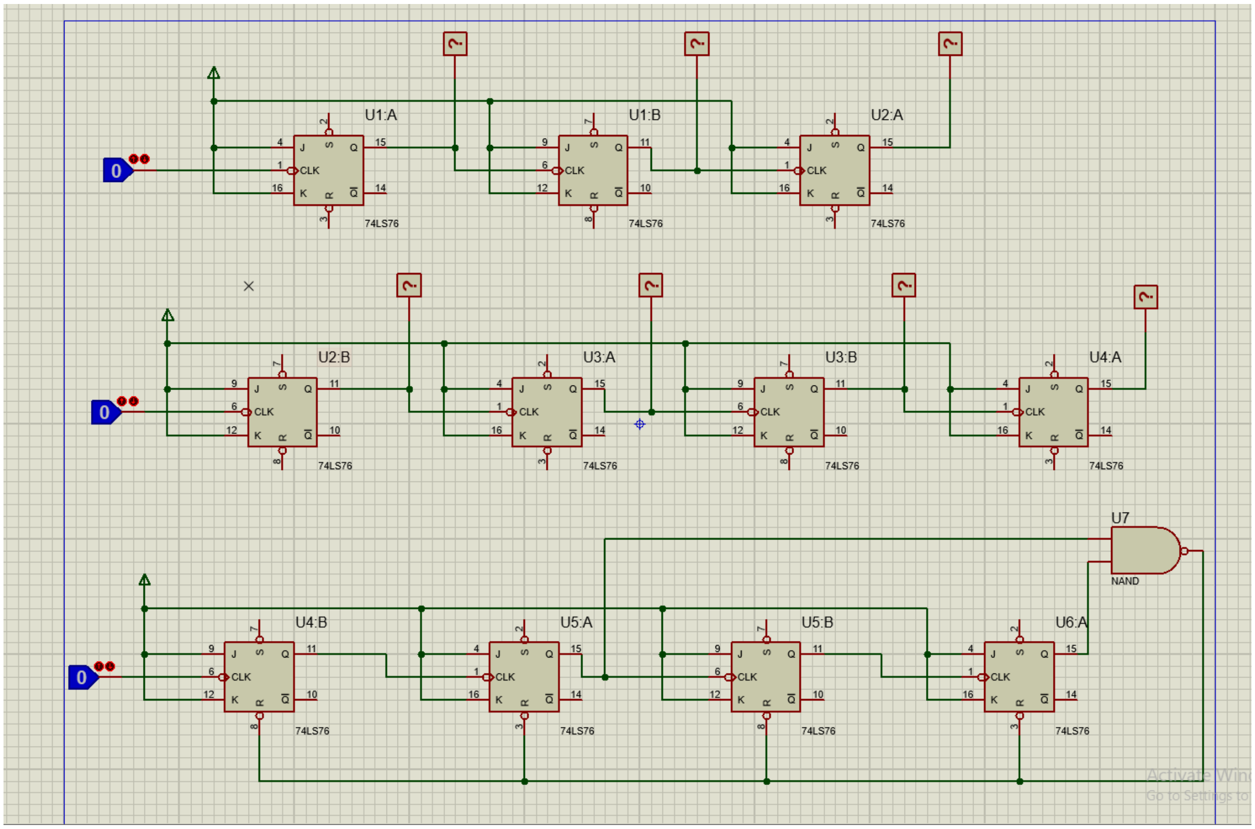
To design and simulate a J-K latch circuit using basic logic gates.

Theory:

A J-K latch is an improved latch that controls the output using J and K inputs and avoids the invalid state of the S-R latch.

Task 16: Counters Circuit

Habiba



Objective:

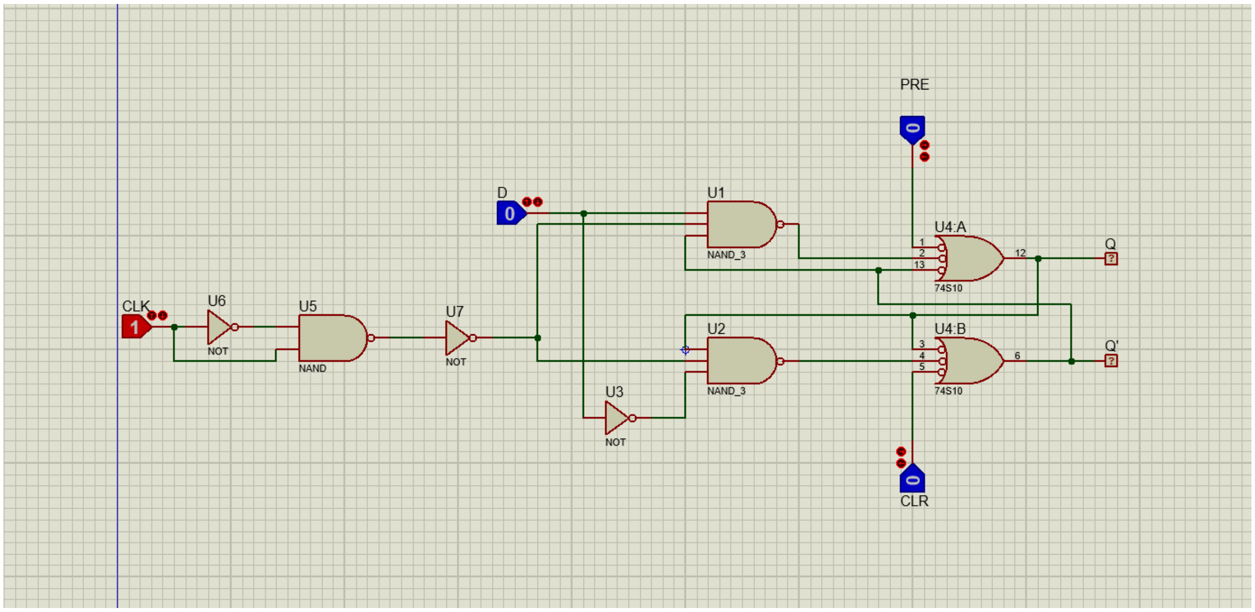
To design and simulate a digital counter circuit using J-K flip-flops.

Theory:

A counter is a sequential logic circuit used to count clock pulses. J-K flip-flops can be connected together to form binary counters, where each flip-flop output represents one bit of the count.

Task 17: D Flip Flop Circuit

Habiba



Objective:

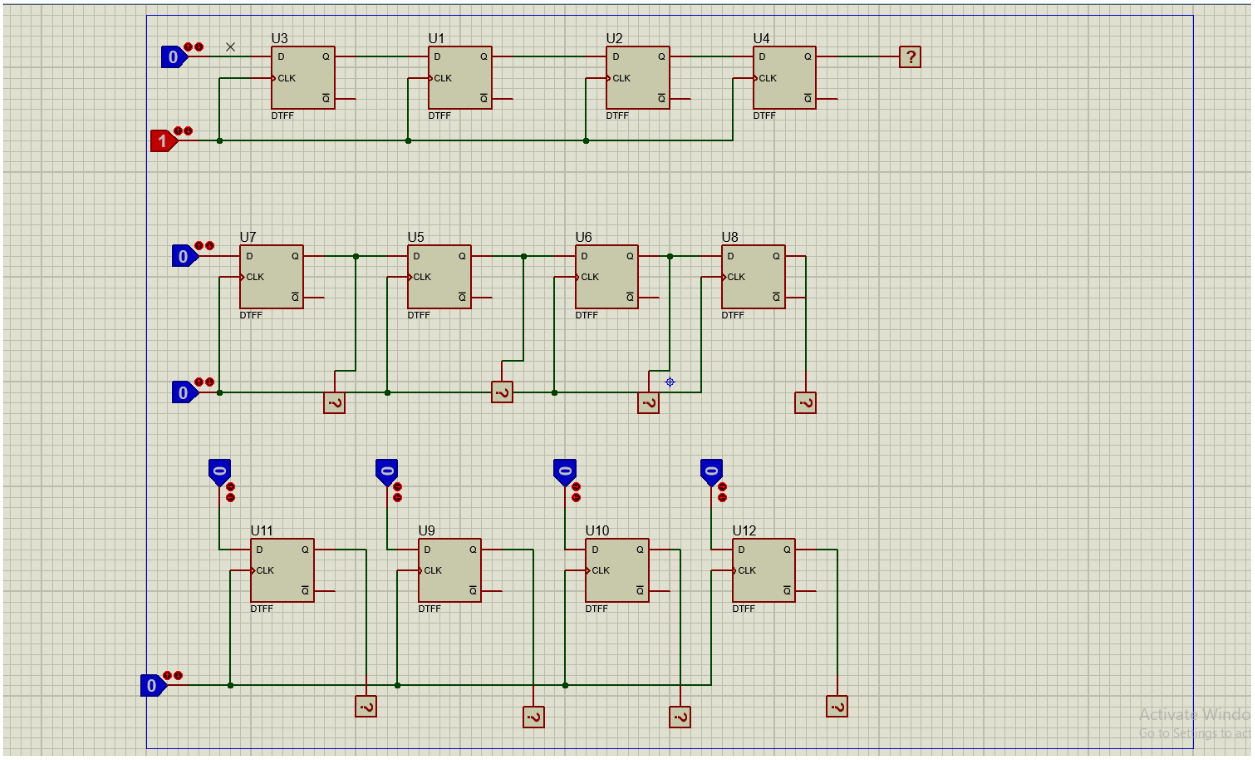
To design and simulate a D flip-flop circuit using logic gates.

Theory:

A D flip-flop is a sequential circuit that stores one bit of data. The output follows the D input only at the active clock edge and holds its state until the next clock pulse.

Task 18: Shift Registers Circuit

Habiba



Objective:

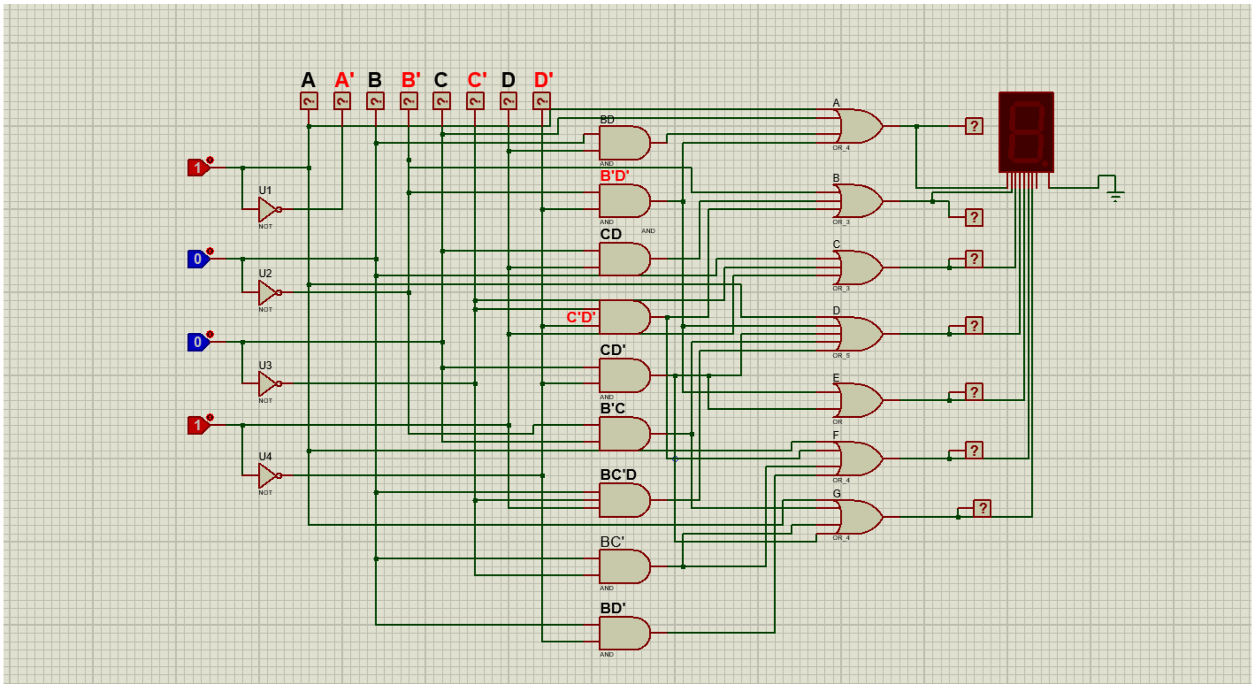
To design and simulate a shift register circuit using D flip-flops.

Theory:

A shift register is a sequential circuit used to store and shift binary data. Each clock pulse moves the data from one flip-flop to the next, allowing serial data transfer and storage.

Task 19: Imp 7-Segment Decoder Circuit

Gerges



Objective:

To design and simulate a 4-bit BCD to 7-segment decoder circuit using basic logic gates.

Theory:

A 7-segment decoder converts a 4-bit binary input into seven output signals. These outputs control the display segments to represent decimal numbers from 0 to 9.